

BALASHOV, Russian Federation

WMO: 341520

Lat: **51.550N**Lon: **43.150E**Elev: **521**StdP: **14.42**Time Zone: **4.00 (E04)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-6.7	-17.2	2.2	-11.0	-12.3	2.9	-6.0	19.6	22.2	18.0	21.9	4.2	30	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	91.6	67.9	87.7	67.2	84.2	66.1	70.9	84.4	69.3	83.0	67.9	80.7	5.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.7	76.3	64.9	93.8	74.6	63.3	88.7	73.7	35.0	84.3	33.7	83.3	32.5	80.7	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	17.1	15.3	13.7	DB	-19.5	94.7	8.1	4.4	-25.3	97.8	-30.0	100.4	-34.6	102.8	-40.5	106.0
(5)				WB	-19.8	75.6	7.9	3.3	-25.5	78.0	-30.1	79.9	-34.6	81.8	-40.4	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	17.6	17.9	28.0	46.6	59.1	66.5	71.2	68.2	57.5	44.7	31.3	21.5
	DBStd	21.58	11.73	12.08	9.12	9.13	8.78	7.63	6.86	7.81	8.02	8.34	9.25	11.63
	HDD50	4450	1005	899	682	168	21	1	0	0	22	207	560	885
	HDD65	8097	1470	1319	1147	552	226	73	19	54	248	629	1010	1350
	CDD50	2376	0	0	0	67	303	496	656	564	247	43	0	0
	CDD65	548	0	0	0	1	43	118	210	153	23	0	0	0
	CDH74	5275	0	0	0	19	421	1077	2026	1525	206	1	0	0
	CDH80	1904	0	0	0	0	104	358	804	601	37	0	0	0
Wind	WSAvg	6.6	7.0	7.7	7.6	7.2	6.5	5.9	5.4	5.5	5.7	6.2	7.1	7.3
Precipitation	PrecAvg	19.7	1.5	1.1	1.1	1.3	1.5	2.2	2.2	1.8	1.7	1.6	2.0	1.8
	PrecMax	29.4	3.3	2.7	2.6	2.8	4.4	6.0	7.4	5.1	4.6	4.7	6.2	3.3
	PrecMin	13.1	0.1	0.0	0.0	0.3	0.1	0.3	0.7	0.0	0.0	0.0	0.2	0.7
	PrecStd	3.8	0.7	0.7	0.6	0.7	1.0	1.4	1.4	1.2	1.3	1.0	1.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	37.2	54.6	76.7	87.1	93.7	96.9	97.3	84.2	70.2	54.2	44.7
		MCWB	39.1	35.2	45.9	56.6	63.7	66.3	67.6	69.7	61.8	56.7	51.3	42.8
	2%	DB	35.6	35.7	47.4	71.7	82.8	88.1	91.8	91.0	79.5	64.9	49.8	39.6
		MCWB	34.4	34.4	42.4	54.7	62.4	66.7	68.5	67.7	61.2	54.4	47.1	38.4
	5%	DB	33.8	34.4	43.0	66.2	78.9	83.8	88.1	86.5	75.6	60.7	46.6	36.5
		MCWB	32.9	33.3	39.0	52.3	61.3	66.0	68.1	67.0	60.0	53.3	44.5	35.5
	10%	DB	32.6	33.2	39.0	62.1	74.5	80.2	84.2	82.3	71.3	57.0	43.2	34.5
		MCWB	31.9	32.2	36.2	50.8	59.3	65.1	67.5	65.1	58.5	51.4	41.5	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	36.0	47.3	59.6	67.2	71.5	74.9	73.3	65.5	60.2	52.0	43.3
		MCDB	40.3	36.5	51.5	72.2	81.1	83.8	84.5	85.5	75.6	65.3	53.7	44.2
	2%	WB	34.9	34.7	43.3	56.5	64.6	69.3	71.4	70.2	63.4	56.9	48.0	38.6
		MCDB	35.3	35.2	47.1	67.1	78.0	81.8	85.2	84.1	74.1	61.5	49.6	39.3
	5%	WB	33.1	33.5	39.3	53.9	62.6	67.9	69.9	68.3	61.6	54.2	44.8	35.7
		MCDB	33.7	34.3	42.4	63.9	75.9	80.1	84.1	82.3	72.2	59.1	46.2	36.3
	10%	WB	31.9	32.4	36.4	51.4	60.5	66.2	68.4	66.6	59.6	51.7	41.7	33.9
		MCDB	32.6	33.3	38.5	60.9	72.3	78.1	81.4	79.4	69.7	56.7	43.2	34.5
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.2	12.1	17.3	18.7	17.8	18.5	19.7	18.1	13.4	9.2	9.1
		MCDBR	6.7	7.7	15.4	24.1	23.1	22.2	22.8	24.1	23.6	19.5	12.7	6.8
		MCWBR	6.8	7.3	10.7	11.7	10.1	9.2	9.1	9.8	11.5	11.7	10.5	6.6
	5% WB	MCDBR	6.9	7.3	14.4	21.7	21.1	19.8	19.8	21.1	20.7	16.5	11.3	6.8
		MCWBR	7.1	7.2	10.4	11.8	10.6	9.6	9.8	9.9	11.4	11.7	10.3	6.8
Clear-Sky Solar Irradiance	taub	0.304	0.339	0.366	0.433	0.399	0.390	0.412	0.432	0.383	0.359	0.335	0.307	
	taud	2.107	2.013	2.064	2.102	2.307	2.360	2.295	2.224	2.343	2.351	2.263	2.138	
	Ebn at Noon	213	236	257	254	271	274	266	252	253	234	203	189	
	Edh at Noon	24	36	42	45	39	37	39	40	32	26	22	20	
All-Sky Solar Radiation	RadAvg	292	624	1082	1370	1816	1930	1922	1637	1105	574	244	162	
	RadStd	72	102	155	116	168	164	162	132	129	92	50	43	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.75	N/A	N/A	+3.24	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	N/A